

Standard Process Inc.

1200 W. Royal Lee Drive
Palmyra, WI 53156

ANALYTICAL REPORT

Received On: 31Oct2024
Reported On: 11Nov2024

Eurofins Sample Code:	468-2024-10310119
Sample Description:	HEMP OIL COMPLEX 8868-3-1-1

Industrial Hemp Cannabinoid Profile	Reference AOAC 2018.11 (modified)	Accreditation	Completed 08Nov2024	Sub 3
Parameter	Result	Theoretical		
Total THC (THC + (THCA x 0.877))	Not Detected %			
Cannabichromene (CBC)	0.070 %			
Cannabichromenic Acid (CBCA)	<0.0017 %			
Cannabicyclol (CBL)	0.005 %			
Cannabidiol (CBD)	2.13 %			
Cannabidiolic Acid (CBDA)	<0.0004 %			
Cannabidivarin (CBDV)	0.010 %			
Cannabidivarinic Acid (CBDVA)	<0.0004 %			
Cannabigerol (CBG)	0.020 %			
Cannabigerolic Acid (CBGA)	<0.0004 %			
Cannabinol (CBN)	0.025 %			
Cannabinolic Acid (CBNA)	<0.0004 %			
Delta 8-THC	<0.0017 %			
Delta 9-THC	<0.0008 %			
Tetrahydrocannabinolic acid (THCA)	<0.0017 %			
Tetrahydrocannabivarin (THCV)	0.003 %			
Tetrahydrocannabivarinic Acid (THCVA)	<0.0004 %			
Total Cannabinoids	2.26 %	>1.32 %		
Total CBD (CBD + (CBDA x 0.877))	2.13 %			
The Hemp Oil Complex (Lot#8868-3-1-1) meets the requirements of <0.3% for Delta 9 THC Levels.				

Glucoraphanin	Reference Internal method	Accreditation	Completed 08Nov2024	Sub 4
Parameter	Result	Theoretical		
Glucoraphanin	4.65 mg/g	>4.39 mg/g		

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Sample Description: HEMP OIL COMPLEX 8868-3-1-1

Pesticides <USP 561> (LC-MS/MS and GC-MS/MS)	Reference USP 561	Accreditation ISO/IEC 17025:2017 A2LA 2993.01	Completed 05Nov2024
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Parameter	Result	Theoretical
Acephate	<0.1 mg/kg	
Alachlor	<0.05 mg/kg	
Aldrin and dieldrin (sum of)	<0.05 mg/kg	
Azinphos-ethyl	<0.1 mg/kg	
Azinphos-methyl	<1 mg/kg	
Bromophos-ethyl	<0.05 mg/kg	
Bromophos-methyl	<0.05 mg/kg	
Bromopropylate	<3 mg/kg	
Chlordane (sum)	<0.05 mg/kg	
Chlorfenvinphos	<0.5 mg/kg	
Chlorpyrifos	<0.2 mg/kg	
Chlorpyrifos-methyl	<0.1 mg/kg	
Chlorthal-dimethyl	<0.01 mg/kg	
Cyfluthrin I-IV	<0.1 mg/kg	
Cyhalothrin, lambda-	<1 mg/kg	
Cypermethrin I-IV	<1 mg/kg	
DDT (sum)	<1.0 mg/kg	
Deltamethrin	<0.5 mg/kg	
Diazinon	<0.5 mg/kg	
Dichlofluanid	<0.1 mg/kg	
Dichlorvos	<1 mg/kg	
Dicofol, o,p' & p,p'-	<0.5 mg/kg	
Dimethoate and omethoate (sum of)	<0.1 mg/kg	
Endosulfan (sum)	<3.0 mg/kg	
Endrin	<0.05 mg/kg	
Ethion	<2 mg/kg	
Etrinfos	<0.05 mg/kg	
Fenchlorophos (sum)	<0.1 mg/kg	
Fenitrothion	<0.5 mg/kg	
Fenpropathrin	<0.03 mg/kg	
Fensulfothion (sum)	<0.05 mg/kg	
Fenthion (sum)	<0.05 mg/kg	

The Hemp Oil Complex (Lot#8868-3-1-1) meets the requirements for pesticides as defined by USP 561.

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Pesticides <USP 561> (LC-MS/MS and GC-MS/MS)	Reference USP 561	Accreditation ISO/IEC 17025:2017 A2LA 2993.01	Completed 05Nov2024
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Parameter	Result	Theoretical
Fenvalerate I&II	<1.5 mg/kg	
Flucythrinate I&II	<0.05 mg/kg	
Fluvalinate, tau- I&II	<0.05 mg/kg	
Fonofos	<0.05 mg/kg	
Heptachlor (sum)	<0.05 mg/kg	
Hexachlorobenzene (HCB)	<0.1 mg/kg	
Hexachlorocyclohexane isomers (other than gamma)	<0.3 mg/kg	
Lindane (gamma-HCH)	<0.6 mg/kg	
Malathion and malaoxon (sum of)	<1 mg/kg	
Mecarbam	<0.05 mg/kg	
Methacrifos	<0.05 mg/kg	
Methamidophos	<0.05 mg/kg	
Methidathion	<0.2 mg/kg	
Methoxychlor	<0.05 mg/kg	
Mirex	<0.01 mg/kg	
Monocrotophos	<0.1 mg/kg	
Parathion-ethyl and paraoxon-ethyl (sum of)	<0.5 mg/kg	
Parathion-methyl and paraoxon-methyl (sum of)	<0.2 mg/kg	
Pendimethalin	<0.1 mg/kg	
Pentachloranisole	<0.01 mg/kg	
Permethrin (sum of isomers)	<1 mg/kg	
Phosalone	<0.1 mg/kg	
Phosmet	<0.05 mg/kg	
Piperonyl butoxide	<3 mg/kg	
Pirimiphos-ethyl	<0.05 mg/kg	
Pirimiphos-methyl (sum)	<4 mg/kg	
Procymidone	<0.1 mg/kg	
Profenofos	<0.1 mg/kg	
Prothiofos	<0.05 mg/kg	
Pyrethrum (total isomers)	<3 mg/kg	
Quinalphos	<0.05 mg/kg	

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Pesticides <USP 561> (LC-MS/MS and GC-MS/MS)		Reference USP 561	Accreditation ISO/IEC 17025:2017 A2LA 2993.01	Completed 05Nov2024	
Parameter	Result	Theoretical			
Quintozene (sum)	<1 mg/kg				
S 421	<0.02 mg/kg				
Tecnazene	<0.05 mg/kg				
Tetradifon	<0.3 mg/kg				
Vinclozolin	<0.4 mg/kg				
The Hemp Oil Complex (Lot#8868-3-1-1) meets the requirements for pesticides as defined by USP 561.					
Product Specification Review - Pesticides		Reference Internal method	Completed 08Nov2024		
Parameter	Result	Theoretical			
Conformance to product specification	Completed				
Heavy Metals As-Cd-Pb-Hg (Low LOQ, ICP-MS)		Reference AOAC 2013.06	Accreditation ISO/IEC 17025:2017 A2LA 2993.01	Completed 08Nov2024	
Parameter	Result	Theoretical			
Arsenic (As)	0.26 ppm	<15 ppm			
Cadmium (Cd)	0.0064 ppm	<5 ppm			
Lead (Pb)	0.0060 ppm	<10 ppm			
Mercury (Hg)	<0.0020 ppm	<2 ppm			
The Hemp Oil Complex (Lot#8868-3-1-1) meets the requirements of heavy metals as defined by USP 2232.					
Fatty Acids-Omega 6 & 3 %W/W		Reference AOCS Ce 2-66 mod., AOCS Ce 1b-89 mod.	Accreditation	Completed 05Nov2024	Sub 1
Parameter	Result	Theoretical			
Fatty Acid Profile, % Weight	Reported as Fatty Acids				
C08:0 Octanoic (Caprylic)	<0.02 %				
C10:0 Decanoic (Capric)	<0.02 %				
Total Saturated Fatty Acids	10.11 %				
C11:0 Undecanoic (Hendecanoic)	<0.02 %				
C12:0 Dodecanoic (Lauric)	<0.02 %				
C14:0 Tetradecanoic (Myristic)	1.13 %				
C14:1 Tetradecenoic (Myristoleic)	0.02 %				
C15:0 Pentadecanoic	0.16 %				
C15:1 Pentadecenoic	<0.02 %				
C16:0 Hexadecanoic (Palmitic)	6.86 %				

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QD089 - Fatty Acids-Omega 6 & 3 %W/W		Reference	Accreditation	Completed
		AOCS Ce 2-66 mod., AOCS Ce 1b-89 mod.		05Nov2024
				Sub 1

Parameter	Result	Theoretical
C16:1 Hexadecenoic (Palmitoleic)	1.81 %	
C16:2 Hexadecadienoic	0.09 %	
C16:3 Hexadecatrienoic	0.04 %	
C16:4 Hexadecatetraenoic	0.07 %	
C17:0 Heptadecanoic (Margaric)	0.21 %	
C17:1 Heptadecenoic (Margaroleic)	0.12 %	
C18:0 Octadecanoic (Stearic)	1.26 %	
C18:1 Octadecenoic (Oleic + isomers)	10.67 %	
C18:2 Octadecadienoic (Linoleic + isomers)	1.92 %	
C18:2 Octadecadienoic Omega 6 (Linoleic)	1.69 %	
C18:3 Octadecatrienoic (Linolenic + isomers)	0.50 %	
C18:3 Octadecatrienoic Omega 3 (Alpha Linolenic)	0.46 %	
C18:3 Octadecatrienoic Omega 6 (Gamma Linolenic)	0.04 %	
C18:4 Octadecatetraenoic Omega 3 (Stearidonic)	0.73 %	
C20:0 Eicosanoic (Arachidic)	0.16 %	
C20:1 Eicosenoic (Gondoic + isomers)	4.24 %	
C20:2 Eicosadienoic Omega 6	0.27 %	
C20:3 Eicosatrienoic	0.22 %	
C20:3 Eicosatrienoic Omega 3	0.16 %	
C20:3 Eicosatrienoic Omega 6	0.06 %	
C20:4 Eicosatetraenoic (Arachidonic + isomers)	1.10 %	
C20:4 Eicosatetraenoic Omega 3	0.52 %	
C20:4 Eicosatetraenoic Omega 6 (Arachidonic)	0.58 %	
C20:5 Eicosapentaenoic Omega 3	6.29 %	>4.83 %
C21:5 Heneicosapentaenoic Omega 3	0.22 %	
C22:0 Docosanoic (Behenic)	0.09 %	
C22:1 Docosenoic (Erucic + isomers)	3.18 %	
C22:2 Docosadienoic Omega 6	0.02 %	
C22:3 Docosatrienoic, Omega 3	<0.02 %	

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Fatty Acids-Omega 6 & 3 %W/W	Reference AOCS Ce 2-66 mod., AOCS Ce 1b-89 mod.	Accreditation	Completed 05Nov2024	Sub 1
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Parameter	Result	Theoretical
C22:4 Docosatetraenoic Omega 6	0.06 %	
C22:5 Docosapentaenoic	1.06 %	
C22:5 Docosapentaenoic Omega 3	0.77 %	
C22:5 Docosapentaenoic Omega 6	0.29 %	
C22:6 Docosahexaenoic Omega 3	11.81 %	>11.41 %
C24:0 Tetracosanoic (Lignoceric)	0.22 %	
C24:1 Tetracosenoic (Nervonic)	0.53 %	
Sum of Omega 3 Isomers	20.98 %	>20.18 %
Sum of Omega 6 Isomers	3.01 %	
Total Fatty Acids Calc.	55.09 %	
Total Fat as Triglycerides	57.47 %	
Total Monounsaturated Fatty Acids	20.57 %	
Total Polyunsaturated Fatty Acids	24.41 %	

Moulds - FDA BAM Chapter 18 mod.	Reference FDA BAM Chapter 18 mod.	Accreditation	Completed 07Nov2024	Sub 2
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Parameter	Result	Theoretical
Mold	< 10 cfu/g	
Yeast	< 10 cfu/g	

Aerobic Plate Count - AOAC 966.23	Reference AOAC 966.23	Accreditation	Completed 07Nov2024	Sub 2
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Parameter	Result	Theoretical
Aerobic Plate Count	10 (est) cfu/g	

Escherichia coli - BAM Chapter 4	Reference FDA BAM Chapter 4	Accreditation	Completed 07Nov2024	Sub 2
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Parameter	Result	Theoretical
E. coli	< 3 MPN/g	

Comments:

Opinion judgments take into account the measurement uncertainty associated with the test results. Regulatory officials may or may not take this uncertainty into account when determining compliance.

The Hemp Oil Complex (Lot#8868-3-1-1) meets the requirements for microbiological levels as defined by USP 2023 (Botanical Extracts).

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Respectfully Submitted,



Victoria Siegel

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ASM Eurofins Analytical Laboratories, Inc.

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